Replacing Phase 1 with Auction-Dynamics MBO

Approach B, measured against the validated deliverables — and a sixth occasion on which an instrument that could not see an effect reported its absence

August 2026

Abstract

Phase 1 (Γ-convergence relaxation by projected gradient descent) costs 22 h–79 h for a completed N=300 solution, and its winner-take-all readout is not a valid partition at that scale, so it is repaired afterwards. This report measures **approach B** — volume-constrained multiphase threshold dynamics [15, 7], with the volume constraint imposed by balanced assignment in the manner of auction dynamics [11] — as an outright *replacement* for Phase 1. Four results. (1) At the pinned Phase 2 campaign, B reaches perimeter **184.4118** and **184.1615** at N=100 (two seeds) against the control's 185.2546, and **319.9428** at N=300 against 323.3192 — from 214–248 s of Phase 1 against 48,132 s and 79,069 s, a 211–319× reduction in the replaced stage. (2) B produces **0 fragmented cells** at both N on its raw labels, where raw PGD at N=300 produces 2 and requires the balanced readout; connectivity is the only one of the three validity gates that has content for B, and it is the one B could have failed. (3) The win is attributable to the MBO dynamics and not to the balanced geodesic initialisation B starts from — which is approach C's first step — by +2.761% against a pre-registered 0.3% threshold. (4) Of thirteen predictions committed before any run, nine held, three missed and one was initially mis-scored. The methodological content is again larger than the technical: a **three-part over-merge instrument was built and detected nothing**, and a per-level claim entered the record having never been measured — the sixth instance of this project's signature artefact.

Provenance

Status	**measured**
Date	2026-08-19
Surface	torus $T(R{=}1.0,\ r{=}0.6)$ throughout
Branch	`feat/phase1-mbo-auction-dynamics`
Code	`src/partition/mbo_auction.py` (the arm); `scripts/run_mbo_arm.py` (driver); `scripts/score_mbo_arm.py` (mechanical scorer); `testing/test_mbo_auction.py` (G1–G8, NC1–NC5b)
Scored runs	`results/arm_mbo_20260819_173411_npart100_V114144_seed84172851`, `results/arm_mbo_20260819_185131_npart100_V114144_seed61803399`, `results/arm_mbo_20260819_192659_npart300_V47488_seed61803399`, `results/arm_init_20260819_181203_npart100_V114144_seed84172851`
Anchors	$N{=}100$: `run_20260709_081548`, Phase 2 best **185.2546144718457** (iterate 20 of 20), *raw* PGD, Phase 1 48132.12 s. $N{=}300$: `run_20260806_123326`, Phase 2 best **323.319247087254** (iterate 18 of 19), PGD *plus* balanced readout, Phase 1 79068.56 s. Both from `timing_profile.yaml total_wall_s`, never `run_time_seconds`.
Pre-registration	commit `9ffdefb`, 2026-08-19 16:53, *before* the first scored run at 17:34:11. Results commits `b414b67`, `c8aea83`, `5bdc947`, `54b937c`; corrections `d4dc66d`, `d4b75d7`.
Figures	`make_figures.py` from the committed `data.yaml`, itself re-derivable by `extract_data.py`.
Versions	Python 3.12.13, numpy 2.4.4, scipy 1.17.1, h5py 3.16.0; macOS 15.3.1 arm64 (Mac mini).
Anchor check	the harness reproduces the $N{=}100$ control's own perimeter *bitwise* (report 07, 0d); no Phase 2 code has changed since.

1 What B is, and what is not claimed for it

Threshold dynamics originates with Merriman, Bence and Osher [15]: alternate short-time diffusion of the phase indicators with a pointwise reassignment, and interfaces move by approximate mean curvature. Esedoḡlu and Otto [7] placed the multiphase case on a variational footing with a Lyapunov functional that Γ-converges to a weighted perimeter — the same limit this project's own energy targets [3, 17, 16]. Jacobs, Merkurjev and Esedoḡlu [11] impose *exact volume constraints* by replacing the pointwise threshold with a balanced assignment solved by Bertsekas' auction algorithm [2]. **None of this is invented here.**

Per mesh level, the arm iterates on a *hard* labelling $\omega : V \to \{0, \dots, N-1\}$:

1. **diffuse**: solve $(M + \tau K)\, y_k = M\chi_k$ for each cell, with one prefactorised LU per level and N back-substitutions;
2. **balanced threshold**: $\omega(i) = \arg\max_k[\, y_{ik} + \psi_k\,]$, with ψ from the shared semi-discrete optimal-transport dual [1, 14, 12] that approach A already ships.

There is never a continuous field whose winner-take-all readout can diverge from what was constrained. B therefore does not *close* the continuous–discrete gap that motivated approaches A–E; the category is absent.

The regime, and the novelty claim we do *not* make. Every published implementation of this machinery we are aware of convolves on a *uniform Cartesian grid via FFT*, on a flat domain, at small N, and for Pólya's *outer* problem — which convex body maximises the length of the shortest partition. That is true of Hu, Liu and Wang [10], whose auction-plus-thresholding machinery is closest to ours but whose domains are $[0,1]^2$ and 3D boxes on $128^2/256^2$ grids, and of Bogosel and Oudet [4], whose capacity-constrained Voronoi construction with > 100 cells is planar. Ours is the *inner* problem: partition a fixed closed genus-1 surface into N equal-area minimal-perimeter cells on an unstructured triangulated mesh, at N in the hundreds. **We phrase this as "we are not aware of", never as "first"**: two web searches and two papers is not a literature review, and surface-remeshing and computer-graphics venues may cover adjacent ground under different vocabulary. Stated as a population claim, the corpus is: two full-text readings plus this project's standing bibliography.

The difference is load-bearing rather than cosmetic. The FEM operator $(M + \tau K)^{-1} M$ is a different object from an FFT convolution, with different failure modes; mesh pinning [18] is ours to manage, and no grid-derived τ heuristic is imported.

2 Result 1 — perimeter

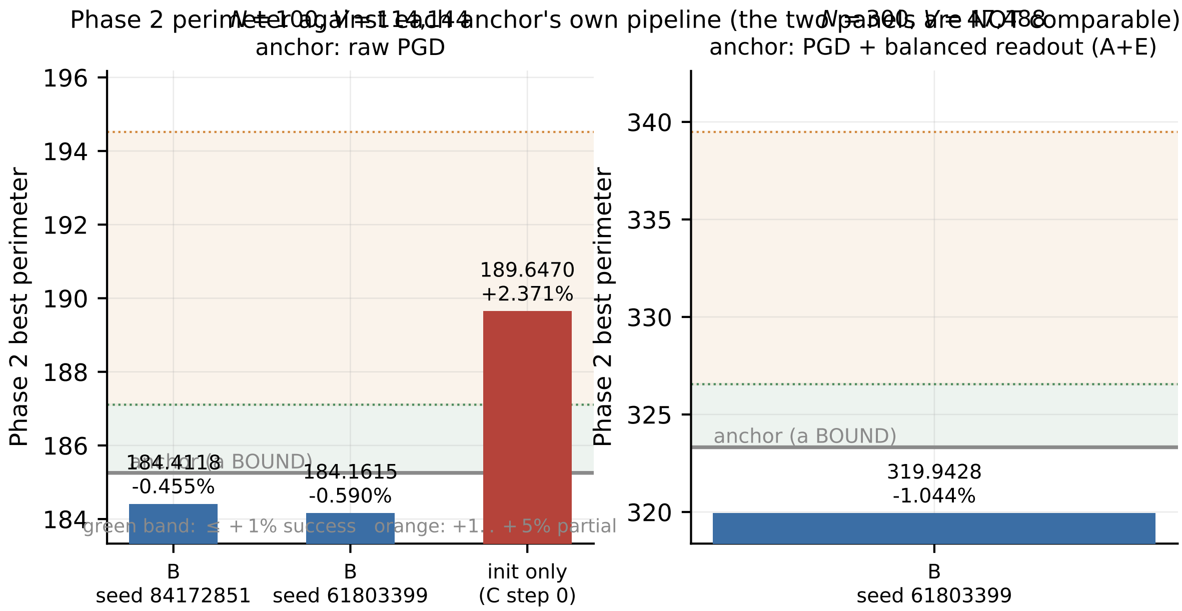

Figure 1: Phase 2 best perimeter against each anchor's own pipeline. **The two panels are not comparable and must never be pooled**: the N=100 anchor is raw PGD, the N=300 anchor is PGD *plus* the balanced readout, because raw PGD is not a valid partition at N=300.

run	Phase 1	speedup	area worst	frag.	Phase 2 best	vs anchor
N=100 s84172851	228 s	210.9×	0.1504%	**0**	184.4117703525215 (18/20)	−**0.455**%
N=100 s61803399	214 s	225.2×	0.1493%	**0**	184.1614647856558 (19/20)	−**0.590**%
N=300 s61803399	248 s	318.8×	1.4172%	**0**	319.9427628182375 (16/19)	−**1.044**%
init only (C step 0)	67 s	—	0.1355%	0	189.6469917457051 (20/20)	+2.37%

Seed spread at N=100 is **0.136**%, which is 3.3× smaller than the margin by which the *worse* of the two seeds beats the control. This is what makes the result scoreable at all: the pre-registration committed that a spread above 1% would mean the primary falsifier could not be evaluated at ±1% and that a single-seed number would be conflating method difference with seed noise.

Six constraints on how these numbers may be stated.

1. **Never pool the two anchors.** They are different baselines. "B beats the best available PGD pipeline at each N, at equal Phase 2 budget" is valid; "B beats PGD by 0.455–1.044%" as a trend in N is not.

2. **The speedup is for the replaced stage only.** End-to-end, including the shared 30–40 min Phase 2, it is $\approx 23\times$ (N=100) and $\approx 31\times$ (N=300). Against the structure-trigger PGD variant (34,967 s), the incumbent-best figure is $\approx 153\times$.

3. **Neither trajectory is a converged optimum.** Both are at the documented migration-cycling plateau. An earlier draft said "B converged where the control had not"; the sixth review resumed the control's own Phase 2 for **20 further iterations and found no improvement** (best over all 40 remains 185.2546 at iterate 20; the extension drifts up to $\approx$ 185.30). B's plateau is simply 0.455% lower. *That extension is the review's measurement and has not been independently reproduced by us.*

4. **PGD's own seed variance is unmeasured.** Every N=100 PGD run across both working directories is seed 84172851, and every N=300 λ=11.5 run is seed 61803399. Each anchor is a single trajectory. "B beats PGD by more than PGD's own seed lottery" is *not* established.

5. **The N=300 margin cannot be decomposed, even in principle.** The readout costs +1.882% in label-boundary length (189.7310 raw → 190.7114 shifted → 193.3024 repaired, the last being what the anchor's Phase 2 consumed). B's raw boundary is 187.2574: shorter than the repaired labels by 3.127% *and* shorter than raw PGD's own invalid labels by 1.304% — the strong form, since imbalance can only make a boundary artificially short. So the margin is not merely "not paying the repair bill". But how much is un-erased repair residue cannot be separated, because **no valid repair-free PGD partition exists at N=300.**

6. N=300 **rests on one Phase-2-scored seed.**

3 Result 2 — validity, and which gate actually has content

"All three gates pass" is true and misleading

An earlier write-up of this result — ours — said B's raw labels pass all three validity gates. Phase 0's own findings say why that is the wrong emphasis:

- **Dormant is vacuous** for any balanced-assignment arm. One-hot labels give peak density 1.0 by construction; the gate cannot fail. The harness already emitted `vacuous_for_arms: True`.
- **Area is near-expected.** Equal area is precisely what B's assignment step *optimises*. Passing means the solver converged; Phase 0 defined it as the *solver-failure detector*, not a validity test.
- **Connectivity is the only gate genuinely at risk.** Nothing in B enforces it, and the source proposal states it is "not per-step guaranteed".

The defensible claim is narrower and still strong: **B produces 0 fragmented cells at N=300, where raw PGD produces 2** (with 10 imbalanced cells, worst 36.15%) — which is why PGD's baseline required the balanced readout and B required nothing.

Re-verified through `testing/check_fragmentation.py`, which shares no code with the driver's reporting: all four arm solutions pass, with **0 cells having more than one raw component including sub-threshold speckle**.

Robustness. Three further N=300 MBO ladders at seeds 27182818, 13131313 and 84172851 give **0 fragmented** in every case, with worst area 1.2531/1.4467/1.3887% and label boundaries 186.08/186.55/186.90 — all *shorter* than the scored seed's 187.2574. Ladder only, no Phase 2; *review-produced, not independently reproduced by us.*

4 Result 3 — attribution: is this B, or is it C's first step?

B is initialised from one balanced C-scores assignment — farthest-point seeds, multi-source Dijkstra, scores $-d^2$, balanced assignment — which is literally approach C's step 0 [13, 1, 6]. If MBO barely improved on it, the number in §2 would be a result about C reported under B's name, and *nothing* in the gate suite would have detected that.

	Phase 1	label boundary	Phase 2 best	censored
init only (C step 0)	67 s	116.1989	189.6469917457051 (20/20)	**yes**
B (MBO)	228 s	107.6014	**184.4117703525215** (18/20)	no
control (PGD)	48,132 s	107.3410	185.2546144718457 (20/20)	plateaued

$$\text{P13} = \frac{189.646992 - 184.411770}{189.646992} = +\mathbf{2.761}\% \qquad \text{against a pre-registered threshold of } 0.3\%.$$

Two disclosures. The measured 2.761% is an **upper bound** on MBO's contribution, because the init arm is censored: its converged value is ≤ 189.6470, and a lower true value shrinks the gap. For attribution to fail the init would have to reach ≈ 185.0 — another $\approx 2.5\%$ on a trajectory gaining ≈ 0.038 per iterate and decelerating. And the init-only control was run on the *finest* mesh, i.e. the *strongest* member of the init family, which biases the test *against* B.

An unflattering free datum on C. C's step 0 plus Phase 2 lands at +2.37% above the PGD control — a PARTIAL verdict under this project's own bands — for 67 s of Phase 1. Real C iterates Lloyd and would improve on this, so it is a *lower bound* on C; but C's starting point is behind both B and PGD, which is directly relevant to whether C is worth building. Note also that C minimises a *quantisation* energy, not perimeter; the two optima coincide only asymptotically and only in the plane [9, 8], and this torus has Gaussian curvature of both signs.

5 The mesh ladder, and τ

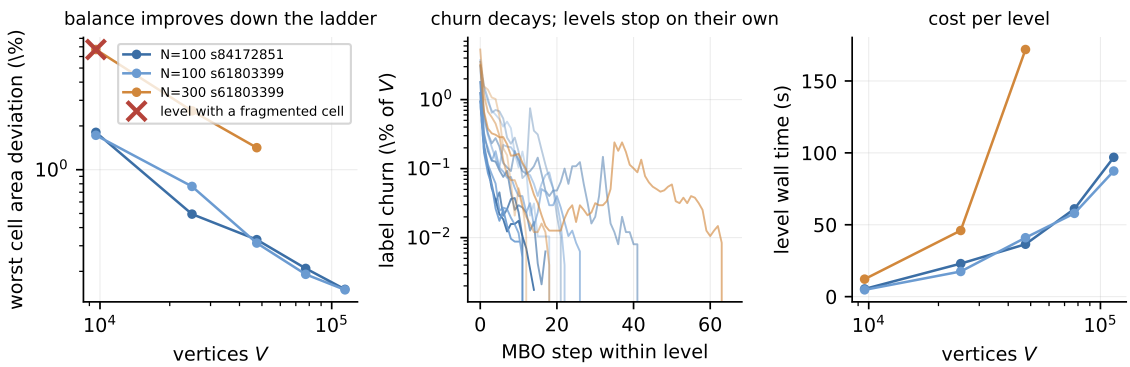

Figure 2: Per-level behaviour. Left: worst cell area deviation improves monotonically down the ladder; the cross marks the one level carrying a fragmented cell. Middle: label churn decays within every level, so levels terminate on their own rather than at the iteration cap. Right: cost per level.

$\tau = \min\big((c\,h_{\mathrm{mean}})^2,\ (\rho R_{\mathrm{cell}})^2\big)$ with $c = 4$ and $\rho = 1.0$, pinned before any run. $c = 4$ rather than $c = 2$: the latter is the freeze regime [18]. The cap is the over-merge side of a two-sided window, and it binds on exactly one level in use, N=300 level 0.

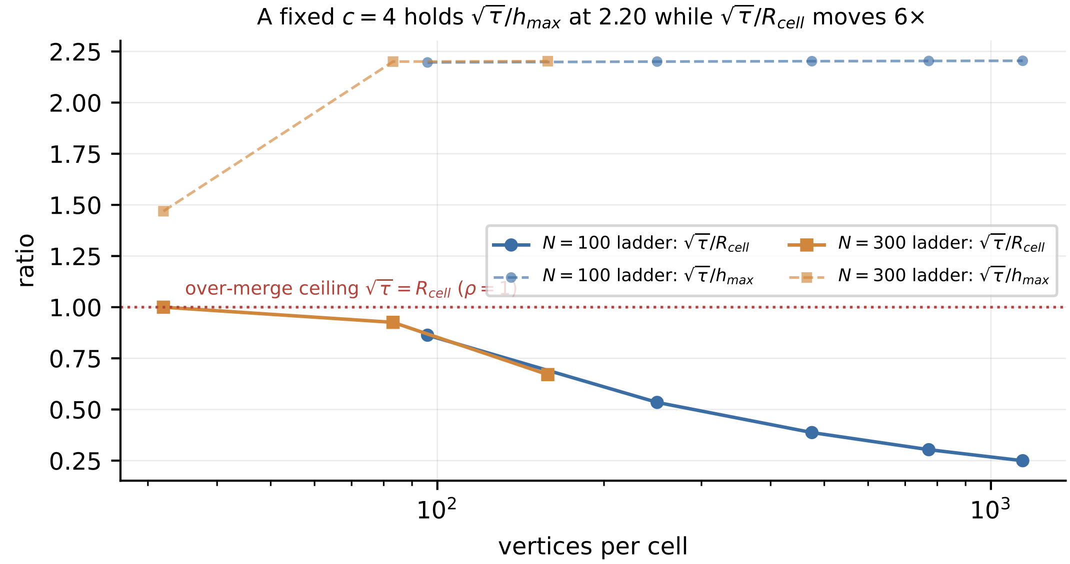

Figure 3: A fixed c does *not* hold the window. $\sqrt{\tau}/h_{\mathrm{max}}$ stays at 2.20 across every level of both ladders while $\sqrt{\tau}/R_{cell}$ moves 6×. Reporting on h_{mean} alone flatters the coarse band: this mesh is 1.81× anisotropic.

6 The headline null: an instrument that detected nothing

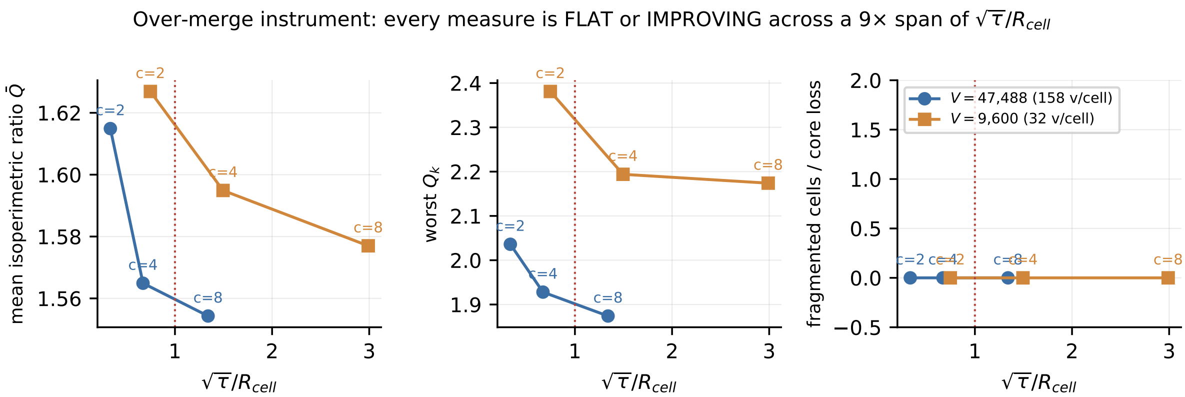

Figure 4: Fragmentation, per-cell isoperimetric ratio $Q_k = P_k^2/(4\pi A_k)$, and core loss, swept over $c \in \{2, 4, 8\}$ at two resolutions. Every measure is flat or *improving*; none registers an over-merge penalty anywhere, including at $\sqrt{\tau}/R_{cell} = 2.99$, three times the pinned cap.

Assignment quality *cannot* see over-merging — the balanced assignment hits its area target by construction, so a smeared, non-local cell scores as well as a compact one. (An earlier draft of the plan cited a healthy assignment-quality fixture as evidence against over-merging; that was a measurement of a different quantity than the claim it supported, and it was withdrawn.) So three geometry instruments were built. Across a 9× span of $\sqrt{\tau}/R_{cell}$ and two levels differing

$5\times$ in resolution, **not one of them registered anything**: core loss and fragmentation are 0 in all six configurations, and compactness moves monotonically the *other* way.

The instruments are demonstrably *responsive* — $\bar{Q}$ moves 3.8%, Q_{worst} 8%, boundary length 1.9% — they simply move toward improvement. What is established is therefore that **no damage is detectable**, *not* that over-merging cannot occur.

> **Standing limit**
>
> The ρ question is **open**, and no future claim about c or ρ may lean on this report's evidence in either direction. The cap was retained because it is pre-registered and because it biases *against* B — it lowers c to 2.68 at N=300 level 0, and c=2 measures worse than c=4. That is a discipline argument, not an empirical one. **Named follow-up:** c=8 showed no penalty on any instrument *and* a 0.33% shorter starting boundary — a third of the entire +1% success bar — so the cap may be costing perimeter for no measured benefit. That is a separate arm, scored through the same harness.

7 Why there is no exactly-monotone energy, and what is gated instead

Esedoğlu–Otto monotonicity is a minorise–maximise argument requiring (i) the same symmetric positive semi-definite form in the energy and in the assignment objective, and (ii) an *exact* assignment. **Both fail here, for independent reasons.** The theorem's object is $MA_\tau = M(M+\tau K)^{-1}M$, symmetric PSD; our step maximises $\langle \chi', Dy\rangle$ with $D = \text{diag}(v)$ *lumped*, and DA_τ is not symmetric. And Jacobs, Merkurjev and Esedoğlu obtain a theorem because Bertsekas' auction is exact, whereas `solve_dual_offsets` is subgradient ascent returning its best iterate. **This is the price of a substitution made deliberately in Phase 0** — where Sinkhorn [5] and auction were both measured and both lost to the incumbent on assignment quality — paid here rather than there.

What survives needs neither concavity nor exactness. Since $\omega'(i) = \arg\max_k[y_{ik}+\psi_k]$, multiplying the pointwise optimality by $v_i > 0$ and summing gives

$$\boxed{\langle \chi', Dy\rangle - \langle \chi, Dy\rangle \;\geq\; \sum_k \psi_k\,(T_k - T'_k)}$$

whose right-hand side is exactly the slack that inexactness costs; it vanishes at perfect balance. Measured slack runs 1.7×10^{-4} to 5.5×10^{-3} — five to seven orders of magnitude above the 10^{-9} tolerance an earlier draft proposed for energy monotonicity, so that tolerance was never defensible. **Real E_τ increases do occur** (4 of 96 active steps at N=300), and gate G4 therefore gates the inequality, whose measured violation is $\leq 6.13\times10^{-8}$, while E_τ is reported as a descriptive trend only.

8 Predictions, and what they caught

	PREDICTED (committed 9ffdefb)	MEASURED	
P1	188.0 (+1.5%), PARTIAL	184.4118 / 184.1615, SUCCESS	**MISS**
P2	spread < 0.5%	0.136%	**HELD**
P3	<= 0.28%, point 0.15%	0.1504 / 0.1493%	**HELD**
P4	<= 3, point 0	0	**HELD**
P5	>= 30x, point 53x	210.9 / 225.2x	**HELD (point missed)**
P6	331.4, 80% CI [323.3, 341.0]	319.9428	**MISS**
P7	<= 2.02%, <= 6 frag, >= 88x	1.4172%, 0, 318.8x	**HELD**
P8	<= 2 of 5 flagged	0, clauses agree	**HELD**
P8'	dE flags L2,L3,L4 (falsifying P8)	dE flat 14-22, flags none	**MISS**
P9	frag not death; worst at L0	1 frag at L0 (295), 0 elsewhere	**HIT**
P10	viol <= 1e-12; excursions <= ceiling	-6.13e-08; 4/96 within ceiling	**HELD**
P12	no abort	20/20, 19/19	**HELD**
P13	>= 0.3%	+2.761%	**HELD**

Figure 5: Thirteen predictions committed in `9ffdefb` at 16:53, before the first scored run at 17:34:11.

P1 and P6 missed, and P1's *reasoning* is refuted. The rationale was that B sees perimeter only through a coarse-τ surrogate while PGD saw the actual energy for 13.4 h. B also *enters* Phase 2 from a 0.243% *longer* boundary than the control and still finishes ahead — a genuine crossover, so the advantage is not initial geometry. That it is combinatorial is a hypothesis consistent with these data, not established by them. P6 missed worse: its 80% interval's lower bound was the anchor itself, so essentially no probability was assigned to B beating PGD + A + E at N=300.

P8′ missed and invalidated its own model. It predicted, from a log-linear interpolation of ΔE/tail in the freeze ratio c/c_{lo}, that levels 2–4 would be flagged pinned. Measured, ΔE/tail is *flat* across the ladder — 15.76, 21.62, 15.11, 14.22, 16.49 against a threshold of 79.25 — and level 4, sitting exactly at $c/c_{\mathrm{lo}} = 0.999$, scores 16.49 where the model demanded ≈ 260. The freeze ratio does not predict pinning here.

9 The sixth artefact — and it is ours

Report 07 catalogued five occasions on which a measurement artefact was read as a result about the method. This is the sixth, and it was produced in this work.

The plan asserted **"0 fragmented at every level"** and, on that basis, scored P9 — which predicted over-merge would appear as fragmentation rather than cell death, worst at level 0 — as UNTESTABLE, "the predicted phenomenon never occurred". **Until the sixth review, `LevelReport` carried no connectivity field at all.** Per-level fragmentation had never been computed. The claim was an inference presented as a result, and it was false.

With the instrument added and the N=300 ladder re-run to a bit-exact reproduction (worst areas 6.7063/2.5394/1.4172%, steps 16/21/66, boundary 187.2574):

level 0: 1 fragmented cell (cell 295). Levels 1 and 2: 0. No cell death.

The phenomenon occurred, at exactly the level P9 named, in exactly the form P9 predicted. **P9 is a hit, not untestable** — "untestable" had absorbed a correct prediction on the strength of a measurement never taken. The stray is healed by level 1's MBO run, which is direct evidence for the mechanism the taxonomy predicted: stray islands are perimeter-inefficient and shrink under curvature flow, the opposite dynamic to the discrete-area trim that manufactured them.

The fix is instrumentation — `run_gates` per level, seconds of cost, now committed — not a resolution to be careful.

10 Threats to validity that no instrument here addresses

1. **PGD's seed variance** (§2, constraint 4). Both anchors are single trajectories; the corpus was enumerated across *both* working directories, since `results/` is gitignored and each worktree has its own.

2. **Over-merging** (§6). Not excluded, merely undetected.

3. **Phase 2 budget as a confound.** B may start from geometry that lets Phase 2 converge *within* a budget PGD cannot use as well. The 40-iteration control extension argues against this, but is review-produced.

4. **Same-machine identity** of the July control and the August arms is consistent with the recorded hardware but is not stamped in run metadata.

5. N=1000 is untouched. Nothing here establishes that B's behaviour extrapolates.

11 Conclusions

1. B replaces Phase 1 at 211–319× less wall time in the replaced stage and reaches a *shorter* Phase 2 perimeter than the best available PGD pipeline at both N, at equal Phase 2 budget.

2. Its raw labels are connected at both N — the one validity gate B could have failed — so the readout and repair stages are unnecessary for B.

3. The advantage is attributable to the MBO dynamics, not to the balanced geodesic initialisation, by +2.761% against a 0.3% threshold.

4. Three pre-registered predictions missed, all in B's favour, and one was initially mis-scored in B's *dis*favour by an instrument that did not exist.

Provenance of the two closest references

Both papers are held at `docs/papers/`. Authors, titles and versions are now **verified against their title pages**: [10] is Shilong Hu, Hao Liu and Dong Wang (v1, 25 May 2024) — an earlier draft of this report guessed two of the three given names wrongly, from a secondary description — and [4] is Beniamin Bogosel and Edouard Oudet (v2, 7 June 2021).

The scope characterisation is now partly checked directly rather than taken on trust: [10] states in its abstract that it treats the *longest* minimal length partition problem by Gaussian convolution with auction dynamics and threshold dynamics, on 128×128 and 256×256 grids in 2D and 3D, and its full text contains **zero** occurrences of “manifold”, “Riemannian” or “torus”; [4] states in its abstract that it seeks the set maximizing the perimeter of the least-perimeter partition and uses capacity-constrained Voronoi diagrams as *initialization*, with 44 occurrences of “convex” and 2 of “manifold”. **What remains second-hand** is the reading of their full numerical sections; the author has read the title pages, abstracts and these keyword counts, not the papers end to end.

Volume and page numbers throughout `docs/math/shared/references.bib` carry an upstream note that they should be checked against published sources before external distribution. That note still stands for the other 28 entries, none of which this report verified.

Related documents

- Plan, pre-registration and full correction history: `docs/plans/PHASE1_BC_REPLACEMENT_PLAN.md`
- Taxonomy: `docs/reference/PHASE1_HIGHN_APPROACHES_ABCDE.md`
- The instrument this report is scored through: `docs/experiments/07-phase0-shared-harness/`
- The gap B makes vacuous: `docs/reference/winner_take_all_partition_gap.md`

References

[1] Franz Aurenhammer, Friedrich Hoffmann, and Boris Aronov. Minkowski-type theorems and least-squares clustering. *Algorithmica*, 20(1):61–76, 1998.

[2] Dimitri P. Bertsekas. The auction algorithm: A distributed relaxation method for the assignment problem. *Annals of Operations Research*, 14(1):105–123, 1988.

[3] Beniamin Bogosel and Édouard Oudet. Partitions of minimal length on manifolds. *Experimental Mathematics*, 26(4):496–508, 2017. YEAR CORRECTED 2026-08-21: this entry read year 2023 with no volume or pages, and its key was `bogosel2023partitions`. Crossref gives print publication 2 October 2017 (online 4 October 2016), Experimental Mathematics 26(4):496–508. This is the foundational reference for the Phase 1 relaxation and Phase 2 refinement methodology.

[4] Beniamin Bogosel and Edouard Oudet. Longest minimal length partitions. *arXiv preprint arXiv:2102.02891v2*, June 2021. Outer problem: numerical evidence that the ball maximizes the perimeter of the least-perimeter partition under volume constraint. Uses capacity-constrained Voronoi diagrams as INITIALIZATION. Verified against `docs/papers/2102.02891v2.pdf` (v2, 7 June 2021); 44 occurrences of “convex”, 2 of “manifold”.

[5] Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. In *Advances in Neural Information Processing Systems*, volume 26, pages 2292–2300, 2013.

[6] Fernando de Goes, Katherine Breeden, Victor Ostromoukhov, and Mathieu Desbrun. Blue noise through optimal transport. *ACM Transactions on Graphics*, 31(6):171:1–171:11, 2012.

[7] Selim Esedoḡlu and Felix Otto. Threshold dynamics for networks with arbitrary surface tensions. *Communications on Pure and Applied Mathematics*, 68(5):808–864, 2015.

[8] Allen Gersho. Asymptotically optimal block quantization. *IEEE Transactions on Information Theory*, 25(4):373–380, 1979.

[9] Thomas C. Hales. The honeycomb conjecture. *Discrete & Computational Geometry*, 25(1):1–22, 2001.

[10] Shilong Hu, Hao Liu, and Dong Wang. Iterative thresholding methods for longest minimal length partitions. *arXiv preprint arXiv:2405.16040v1*, May 2024. Auction dynamics + threshold dynamics for the OUTER (Pólya) problem; Gaussian convolution on 128×128 and 256×256 Cartesian grids, 2D and 3D. Authors verified against the title page of `docs/papers/2405.16040v1.pdf`; zero occurrences of manifold/Riemannian/torus in the full text.

[11] Matt Jacobs, Ekaterina Merkurjev, and Selim Esedoḡlu. Auction dynamics: A volume constrained MBO scheme. *Journal of Computational Physics*, 354:288–310, 2018. AUTHORS CORRECTED 2026-08-21, verified against dblp (journals/jcphy/JacobsME18) and ScienceDirect. This entry previously read "Jacobs, Kim and Léger", an error that originated in the archived A–E proposal (`docs/reference/phase1_highn_proposal_ABCDE_original.md`) and propagated into seven documents before being checked. Volume, pages and year were correct throughout. Note Esedoḡlu is an author of both this and `esedoglu2015threshold`.

[12] Jun Kitagawa, Quentin Mérigot, and Boris Thibert. Convergence of a Newton algorithm for semi-discrete optimal transport. *Journal of the European Mathematical Society*, 21(9):2603–2651, 2019.

[13] Stuart P. Lloyd. Least squares quantization in PCM. *IEEE Transactions on Information Theory*, 28(2):129–137, 1982.

[14] Quentin Mérigot. A multiscale approach to optimal transport. *Computer Graphics Forum*, 30(5):1583–1592, 2011.

[15] Barry Merriman, James K. Bence, and Stanley J. Osher. Motion of multiple junctions: A level set approach. *Journal of Computational Physics*, 112(2):334–363, 1994.

[16] Luciano Modica. The gradient theory of phase transitions and the minimal interface criterion. *Archive for Rational Mechanics and Analysis*, 98(2):123–142, 1987. Classical Γ-convergence result for the Modica–Mortola functional; source of the interface constant $c_W = 2\int_0^1 \sqrt{W}$ used in the Phase 1 energy.

[17] Luciano Modica and Stefano Mortola. Un esempio di γ-convergenza. *Bollettino dell'Unione Matematica Italiana B (5)*, 14:285–299, 1977. The original Γ-convergence example behind the Phase 1 relaxation energy.

[18] Yves van Gennip, Nestor Guillen, Braxton Osting, and Andrea L. Bertozzi. Mean curvature, threshold dynamics, and phase field theory on finite graphs. *Milan Journal of Mathematics*, 82(1):3–65, 2014.